popquiz equilibrium

This is a preview of the draft version of the quiz

Started: Apr 29 at 7:57pm

Quiz Instructions

Question 1 25 pts

image.png

hint: see picture below. The magnitude of the tension is the same in both of the strings. T1=T2=T

1) The Sum(up) = sum(down) so Ty + Ty = weight.

solve for Ty. then

2) Use your trig skills to find T using : Ty = T x ....(40) is .... sine? or cosine? or tan ?

solve for T.

image.png

- 389N
- 778N
- 500N
- 326N

Question 2 25 pts

image.png

What is true?

- T1=T2= 250N
- T2=250N, T1=0

- ○ T1=T2=500N
- ○ T2=500N, T1=0N

Question 3 25 pts

image.png

What is the normal force?

hint: the pull P has 2 components shown below. Px and Py.

sum(up)=sum(down)

Equ1 N + Py = 400

Use your trig skill to find Py = P x ... (30) with P =100N. Is ... cosine ? sine? tangent ?

plug Py in equation1 and solve for N.

- ○ 100N
- ○ 350N
- ○ 400N
- ○ 313N

Question 4 25 pts

The crate at the origin is in equilibrium and is pulled by 3 strings. The 2 blue strings make an angle of 30 degrees . The tensions in the blue string are T1 and T2 . T1=T2=T in magnitude = 10N.

What is the tension in the orange string? (T2)

hint: sum(right) = sum(left) so T3 = Tx + Tx use your trig skills

- ◯ 20N
- ◯ 5N
- ◯ 17.3N
- ◯ 10N

Not saved Submit Quiz